21-22秋线性代数期末(回忆版)

1. Label the following statements as True or False. Along with your answer, provide an informal proof, ~~con~~ counterexample, or other explanation.
   (a) The sum of two positive operators on a finite-dimensional complex inner product space is positive.
   (b) Let V be a 5-dimensional vector space and $T \in \mathcal{L}(V)$. Then there exists a 3-dimensional subspace U of V invariant under T.
   (c) Any polynomial of degree n with leading coefficients $(-1)^n$ is the characteristic polynomial of some linear operator.
   (d) If x, y and z are vectors in an inner product space such that $\langle x,y\rangle = \langle x,z\rangle$, then $y=z$.
   (e) Every normal operator is ~~dig~~ diagonalizable.
2. Suppose $T \in \mathcal{L}(R^3)$ is defined by
$$T(x_1,x_2,x_3) = (2x_1, x_2-x_3, x_2+x_3)$$
   (a) Determine the eigenspace of T corresponding to each eigenvalue.
   (b) Find the Jordan form and a Jordan basis of T.
   (c) Find the minimal polynomial of T.
   (d) Find the trace of T, trace T.
   (e) Find the determinant of T, det T.
3. Suppose V is a finite-dimensional inner product space, $T \in \mathcal{L}(V)$ is normal, and U is a subspace of V that is invariant under T. Show that $U^{\perp}$ is invariant under T.
4. Let T be a linear operator on a finite-dimensional vector space V, and let v be a nonzero vector in V. The subspace
$$U = \text{span}(\{v, Tv, T^2v, \cdots\})$$
   is called the T-cyclic subspace of V generated by v.
   (a) Show that U is a finit-dimensional invariant subspace of V.
   (b) Let $k = \dim U$. Show that $\{v, Tv, T^2v, \cdots, T^{k-1}v\}$ is a basis ~~of~~ for U.
   (c) If $a_0v + a_1Tv + a_2T^2v + \cdots + a_{k-1}T^{k-1}(v) + T^k(v) = 0$, show that the characteristic polynomial of $T|_U$ is $f(t) = (-1)^k(a_0 + a_1t + \cdots + a_{k-1}t^{k-1} + t^k)$.
   (d) Let $g(t)$ be the characteristic polynomial of T, show that $g(T)=0$, where 0 is the zero operator. That is, T "satisfies" it characteristic equation.
5. If $F=C$, show that T is an isometry if and only if T is normal and $|\lambda|=1$ for every eigenvalue λ of T.

6. Let $P_2(R)$ and $P_1(R)$ be the polynomial spaces with inner products defined by

$$\langle f,g\rangle = \int_{-1}^{1} f(x)g(x)\,dx\,,\quad f,g\in P_2(R)$$

Let $T: P_2(R)\to P_1(R)$ be the linear operator defined by

$$T(f(x)) = f'(x)$$

(a) Find orthonormal bases $\{v_1,v_2,v_3\}$ for $P_2(R)$ and $\{u_1,u_2\}$ for $P_1(R)$.

(b) Find $p\in P_1(R)$ that makes $\int_{-1}^{1}|x^5-p(x)|^2\,dx$ as small as possible.

(c) Find the singular values σ_1,σ_2 of T such that $T(v_i)=\sigma_i u_i$, $i=1,2$, and $T(v_3)=0$.

7. Let V be a real inner product space. A function $f: V\to V$ is called a rigid motion if

$$\|f(x)-f(y)\| = \|x-y\|$$

for all $x,y\in V$. For example, any isometry on a finite-dimensional real inner product space is a rigid motion. Another class of rigid motions are the translations. A function $g: V\to V$, where V is a real inner product space, is called a translation if there exists a vector $v_0\in V$ such that $g(v)=v+v_0$ for all $v\in V$. Let $f: V\to V$ be a rigid motion on a finite-dimensional real inner product space V, show that there exists a unique isometry T on V and a unique translation g on V such that $f=g\circ T$.